

- Q.29 Write five points of distinction between weir and barrage. (CO8)
- Q.30 Write short note on silt excluder and silt ejector. (CO8)
- Q.31 Explain cross drainage works. (CO9)
- Q.32 Write short note on Aqueduct. (CO9)
- Q.33 What are the functions of river training works? Explain. (CO11)
- Q.34 What is cut-off and its advantages. Explain. (CO11)
- Q.35 Write any five causes of water logging. (CO12)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define the following terms related to tube wells:
- Water table
 - Radius of influence
 - Yield of a well
 - Confined aquifer
- Q.37 a) What is canal lining? Describe its advantage and disadvantages. (Co5)
- b) Describe the classification of canals.
- Q.38 Explain the following :
- Lift irrigation
 - Culturable commanded area.
 - Necessity of irrigation
 - Unlined canals.

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**4th Sem / Civil, Brick Tech., Constr, Mgmt.,
Highway Engg.**

Subject:- Irrigation Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The irrigation system in which water available throughout the year is (CO1)
- Perennial
 - Non perennial
 - Both A and B
 - None of the above
- Q.2 The irrigation projects which cover an area of more than 10,000 ha is (CO1)
- Minor irrigation projects
 - Major irrigation projects
 - Medium irrigation projects
 - None of the above
- Q.3 Which one of the following is not a method of surface irrigation. (CO4)
- Free flooding
 - Check flooding
 - Boarder flooding
 - Drip irrigation
- Q.4 The vertical distance between designed F.S.L and bank level is (CO5)
- Side slope
 - Berms
 - Free board
 - None of the above

Q.5 _____ are the gaps created in the canal banks. (CO5)

- a) Spoil bank b) Dowel
- c) Canal breaches d) Borrow pit

Q.6 _____ Zoe lies between the ground surface and the water table. (CO6)

- a) Zone of saturation b) Ground water zone
- c) Zone of aeration d) All of the above

Q.7 A geological formation which permits storage of water but not allows to transmit is known as (CO6)

- a) Aquifer b) Aquifuge
- c) Aquiclude d) All of the above

Q.8 _____ allow disposal of surplus water from upstream to downstream. (CO7)

- a) Free board b) Inspection road
- c) Spillway d) Side slope

Q.9 Dams which are constructed for the purpose of diverting water of the river into off-taking canal is (CO7)

- a) Storage dams b) Diversion dams
- c) Detention dams d) Overflow dams

Q.10 When a canal is carried under a natural drainage the structure is (CO10)

- a) Super passage b) Level crossing
- c) Divide wall d) Aqueduct

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define irrigation. (CO1)

Q.12 Define drip irrigation. (CO4)

Q.13 Define check flooding method of irrigation. (CO4)

Q.14 Define counter berm. (CO5)

Q.15 Define Aquifuse. (CO6)

Q.16 Describe specific yield. (CO6)

Q.17 Define earthen dam. (CO7)

Q.18 Name various types of cross drainage works. (CO9)

Q.19 Define cross regulator. (CO10)

Q.20 Write causes of water-logging. (CO12)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe benefits and ill-effects of irrigation. (CO1)

Q.22 Define duty and state the factors that affect duty. (CO2)

Q.23 What is run off? Explain also the factors affecting run-off. (CO2)

Q.24 Define run-off and write down the factors affecting run-off. (CO3)

Q.25 Explain unit hydrograph. (CO3)

Q.26 Describe a brief note on yield of a tubewell. (CO6)

Q.27 Discuss the cause of failure of earth dam. (CO7)

Q.28 Explain various methods of construction of earth dam. (CO7)